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CS-499 Module 4 Milestone Three

Artifact Enhancement 2: Algorithms and Data Structure

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The artifact I selected for Milestone Three was the Angular administrative trip listing feature from my CS-465 Travlr application. I selected this artifact because it manages a collection of trip objects and originally displayed all trips without any advanced processing or organization. The trip listing provided an ideal opportunity to demonstrate algorithms and data structures concepts by enhancing the way data is searched, sorted, and displayed. Because the application already contained a working CRUD implementation, I was able to focus specifically on improving how data is processed and presented rather than creating entirely new functionality.

The original implementation loaded trip data from the backend and displayed all results directly to the user. To improve this artifact, I redesigned the trip listing logic into a data-processing pipeline that transforms the trip collection before rendering it on the screen. I implemented client-side search, sorting, and pagination functionality. Search was implemented using linear traversal of the trip collection, allowing users to search across multiple fields including trip code, trip name, resort name, and description. To improve consistency and usability, search terms and trip data were normalized using lowercase string comparisons.

I also implemented comparator-based sorting algorithms that allow users to sort trips by multiple criteria, including alphabetical order, trip price, and start date. Different comparison techniques were required depending on the data type being processed. String values were sorted using locale-based comparisons, numeric values were sorted using numerical comparisons, and dates were converted to timestamps before comparison. This enhancement improved the usability of the application while demonstrating practical applications of sorting algorithms.

To further enhance the trip listing, I implemented client-side pagination. Rather than displaying all trips simultaneously, the application now calculates page boundaries and displays only a subset of records at a time. Pagination was implemented using index calculations and

array slicing operations. This approach improves the user experience while demonstrating data partitioning techniques commonly used when working with collections of objects.

This enhancement aligns with several CS-499 course outcomes. It demonstrates the ability to apply computer science principles and software development practices to improve an existing application. The project required analyzing a collection of data, selecting appropriate algorithms for searching and sorting, and implementing efficient techniques for managing and displaying information. The enhancement also demonstrates the ability to evaluate and improve software design through the use of structured data-processing logic and reusable methods.

Through this enhancement, I gained a deeper understanding of how algorithms directly impact application functionality and user experience. While I had previously used built-in array methods, this project helped me better understand the underlying concepts behind linear search, comparator-based sorting, and pagination through index calculations. One challenge I encountered was ensuring that search, sorting, and pagination worked together correctly. Because each operation modifies the collection before it is displayed, I needed to carefully design the processing sequence so that the final results remained accurate and consistent. Another challenge involved handling different data types during sorting, particularly dates and numeric values, which required different comparison approaches than string data. Successfully completing these enhancements strengthened both my understanding of algorithms and my ability to apply data structures concepts within a real-world web application.